

PROJECT DOCUMENTATION

Master of Science — Data Analytics

AI Voice-Assisted Exam System

An intelligent, speech-enabled online examination platform that reads questions aloud, captures spoken answers, evaluates them automatically, and generates scores, certificates and leaderboards.

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1. Abstract

The **AI Voice-Assisted Exam System** is an online examination platform that makes assessments accessible, hands-free and interactive by combining conventional web technology with speech recognition and speech synthesis. Instead of only reading questions on screen, the system speaks each question aloud to the candidate and listens to the answer spoken back. The captured response is transcribed, compared against the stored answer using text-similarity techniques, and marked automatically. This design assists visually-impaired candidates, reduces typing effort, and speeds up the assessment cycle.

The platform serves two classes of users. **Candidates** register, authenticate, take a voice-driven exam, and immediately receive their score, a downloadable certificate and a place on the leaderboard. **Administrators** manage the question bank and user accounts and review results. All information — user credentials, question bank, per-question results and aggregated scores — is persisted in a relational database. This document presents the complete design of the system: its objectives, requirements, technology stack, architecture, data-flow diagrams at Levels 0–2, the entity-relationship model, the database dictionary, module descriptions, testing, and directions for future work.

2. Introduction

2.1 Overview

Traditional online examination systems are entirely visual and keyboard-driven: the candidate reads a question and types or clicks an answer. This excludes candidates with visual impairment or limited typing ability and offers little interactivity. The AI Voice-Assisted Exam System reimagines the exam as a spoken conversation. Using the browser's Web Speech API together with server-side speech libraries, the platform converts questions to speech (text-to-speech) and converts the candidate's spoken answer back to text (speech-to-text) for automatic evaluation.

2.2 Problem Statement

Existing examination portals depend on manual reading and typing, are difficult for differently-abled users, and often require human effort to evaluate descriptive responses. There is a need for an accessible, automated system that can conduct an exam through voice, evaluate answers objectively, and deliver results instantly without manual intervention.

2.3 Motivation

The motivation for this project is threefold: to improve **accessibility** so that candidates who cannot easily read or type can still take exams independently; to introduce **automation** so that answer evaluation, scoring and certification happen without examiner effort; and to demonstrate the practical application of **data-analytics and natural-language techniques** — speech transcription and text-similarity matching — inside a real, database-backed web application.

3. Objectives

The principal objectives of the AI Voice-Assisted Exam System are:

- To provide a secure registration and login mechanism for candidates and administrators.
- To deliver examination questions through **voice output** (text-to-speech) and accept answers through **voice input** (speech-to-text).

- To evaluate spoken answers automatically by comparing the transcript with the stored answer.
- To calculate and store scores, and to generate certificates for successful candidates.
- To maintain a leaderboard that ranks candidates by performance.
- To give administrators full control over the question bank and user records.
- To persist all data reliably in a relational database and return results to the candidate in real time.

4. Scope of the Project

The system covers the complete lifecycle of a voice-based online examination: user onboarding, authentication, question delivery by voice, answer capture and transcription, automatic evaluation, scoring, certificate generation and leaderboard ranking, together with an administrative back-office for managing questions and users. It is delivered as a web application accessible from any modern browser that supports the Web Speech API.

The current scope focuses on objective questions whose answers can be matched against a stored reference string. Proctoring (camera/identity monitoring), payment, and fully descriptive essay grading are considered out of scope for this version and are listed under future enhancements.

5. Existing System vs. Proposed System

5.1 Existing System

Conventional online examination portals present questions on screen and require the candidate to type or select answers. Their main limitations are:

- Not accessible to visually-impaired or motor-impaired candidates.
- Purely visual and non-interactive; no audio delivery of questions.
- Descriptive answers often need manual evaluation, delaying results.
- Limited automation of scoring, certification and ranking.

5.2 Proposed System

The proposed AI Voice-Assisted Exam System addresses these gaps by adding a voice layer and automatic evaluation over a robust database-backed application. Its advantages are:

- Questions are **spoken aloud**; answers are given by **voice**, enabling hands-free, accessible exams.
- Answers are transcribed and **evaluated automatically** using text-similarity matching.
- Scores, certificates and leaderboard positions are generated **instantly**.
- A clean separation between candidate and administrator roles improves security and maintainability.
- All artefacts are stored in a relational database for auditability and analytics.

6. System Requirements

6.1 Hardware Requirements

Component	Minimum Specification
Processor	Dual-core 2.0 GHz (quad-core recommended)
Memory (RAM)	4 GB minimum, 8 GB recommended

Component	Minimum Specification
Storage	500 MB free for the application and database
Peripherals	Microphone and speakers / headset (required for voice)
Network	Stable broadband internet connection

6.2 Software Requirements

Component	Specification
Operating System	Windows 10/11, Linux or macOS
Web Browser	Google Chrome / Edge (Web Speech API support)
Backend Runtime	Python 3.9 or higher
Web Framework	Flask
Database	MySQL 8.x
IDE / Tools	VS Code, Git, MySQL Workbench

6.3 Functional Requirements

- The system shall allow a new candidate to register and shall reject duplicate usernames.
- The system shall authenticate users and maintain a secure session.
- The system shall read each question aloud and capture the candidate's spoken answer.
- The system shall transcribe the answer and evaluate its correctness automatically.
- The system shall compute the total score and persist per-question results.
- The system shall generate a certificate and update the leaderboard on completion.
- The administrator shall be able to add, modify and delete questions and manage users.

6.4 Non-Functional Requirements

- **Usability:** simple, voice-first interface usable with minimal training.
- **Accessibility:** operable by visually- and motor-impaired candidates.
- **Performance:** question delivery and evaluation respond within a couple of seconds.
- **Security:** passwords are hashed; roles separate candidate and admin privileges.
- **Reliability & Availability:** results are stored durably; the app degrades gracefully if voice is unavailable.
- **Maintainability & Scalability:** modular design and a normalized schema support growth.

7. Technology Stack

The system follows a classic three-tier architecture — a browser-based presentation tier, a Python/Flask application tier, and a MySQL data tier — augmented with a voice layer and a lightweight NLP evaluation component. The reference stack implied by the design is:

Layer	Technology	Purpose in the system
Presentation	HTML5, CSS3, Bootstrap 5	Responsive exam & admin interface
Client logic	JavaScript (ES6), AJAX	UI behaviour and asynchronous calls to the server

Layer	Technology	Purpose in the system
Voice interface	Web Speech API (SpeechSynthesis & SpeechRecognition)	Reads questions aloud and captures spoken answers in the browser
Application	Python 3, Flask	Routing, business logic and REST endpoints
Speech (server)	SpeechRecognition, pyttsx3 / gTTS	Server-side speech-to-text and text-to-speech fallback
Evaluation	NLTK, difflib / fuzzy matching	Compares the spoken answer with the stored answer
Database	MySQL 8.x	Stores users, questions, results and scores
DB connector	mysql-connector-python / SQLAlchemy	Database access layer
Security	Werkzeug password hashing, Flask sessions	Authentication and session management
Deployment	Gunicorn + Nginx (production)	Application hosting
Tooling	VS Code, Git & GitHub, MySQL Workbench	Development, version control and DB design

8. System Architecture

At a high level the browser presents the exam and handles voice input/output through the Web Speech API. Spoken answers are transcribed and sent to the Flask application, which authenticates the user, fetches questions, evaluates answers, computes scores and reads from and writes to the MySQL database. The context diagram below (DFD Level 0) captures this interaction between the two external actors — **Users** and **Admin** — the central process, and the data store.

DFD Level 0 — Context Diagram

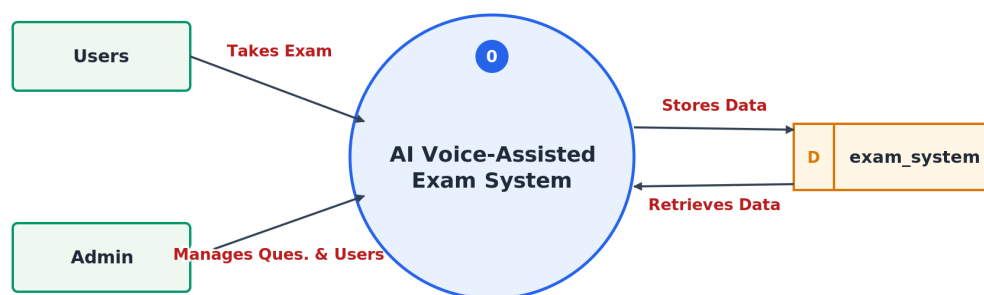


Figure 1: Context Diagram (DFD Level 0)

9. Data Flow Diagrams

Data-flow diagrams model how information moves through the system. The notation used throughout this document is summarised below.

Symbol	Element	Meaning
Circle / bubble	Process	A transformation of data (e.g., User Authentication)
Rectangle	External entity	A source or sink outside the system (User, Admin)
Open-ended box	Data store	A repository of data (e.g., exam_system, exam_data)
Arrow	Data flow	Movement of data between elements

9.1 Level 1 DFD

The Level 1 diagram decomposes the single context process into four major processes — **User Authentication**, **Exam Process**, **Admin Management** and **Result Processing** — and shows the three data stores they use. A candidate logs in, starts the exam, submits answers, and views results; the administrator manages users and questions.

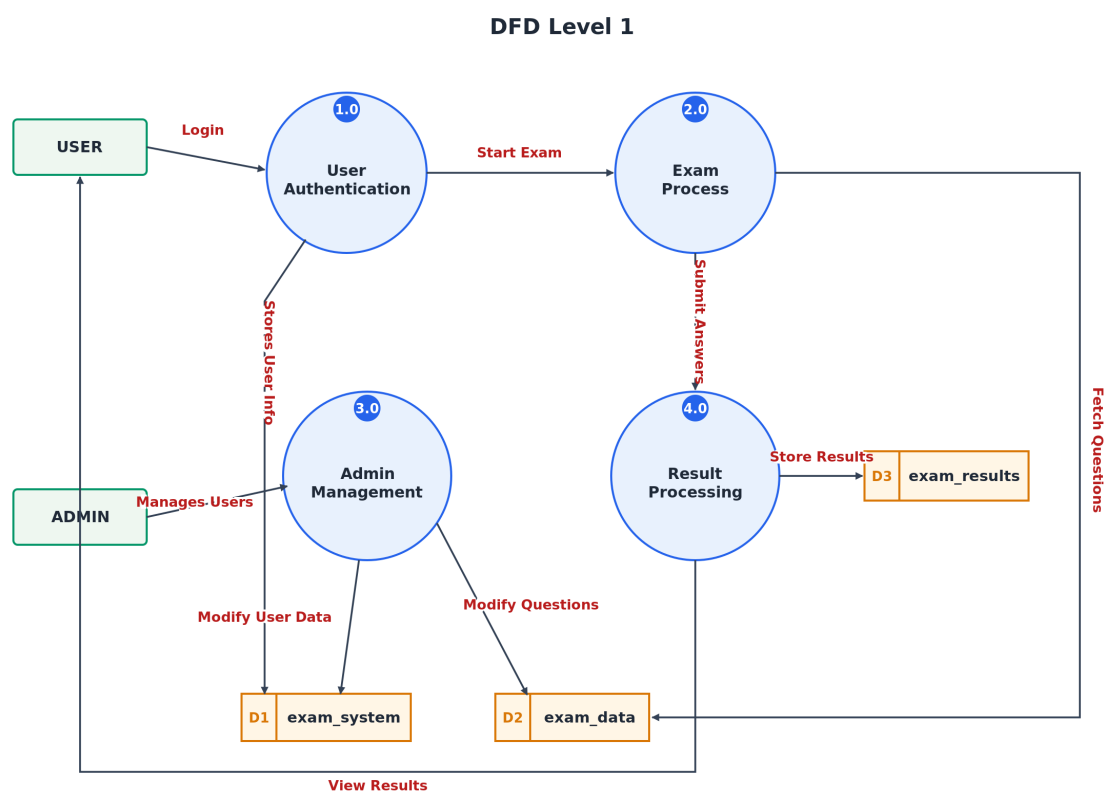


Figure 2: Level 1 Data Flow Diagram

9.2 Level 2 DFD

The Level 2 diagram expands the processes into finer sub-processes: user registration and login, fetching exam questions, processing voice answers, evaluating and calculating the score, and the post-exam steps of generating a certificate and updating the leaderboard. The 'Evaluate Answer' flow feeds evaluated responses back into the scoring processes, and results are returned to the user.

DFD Level 2

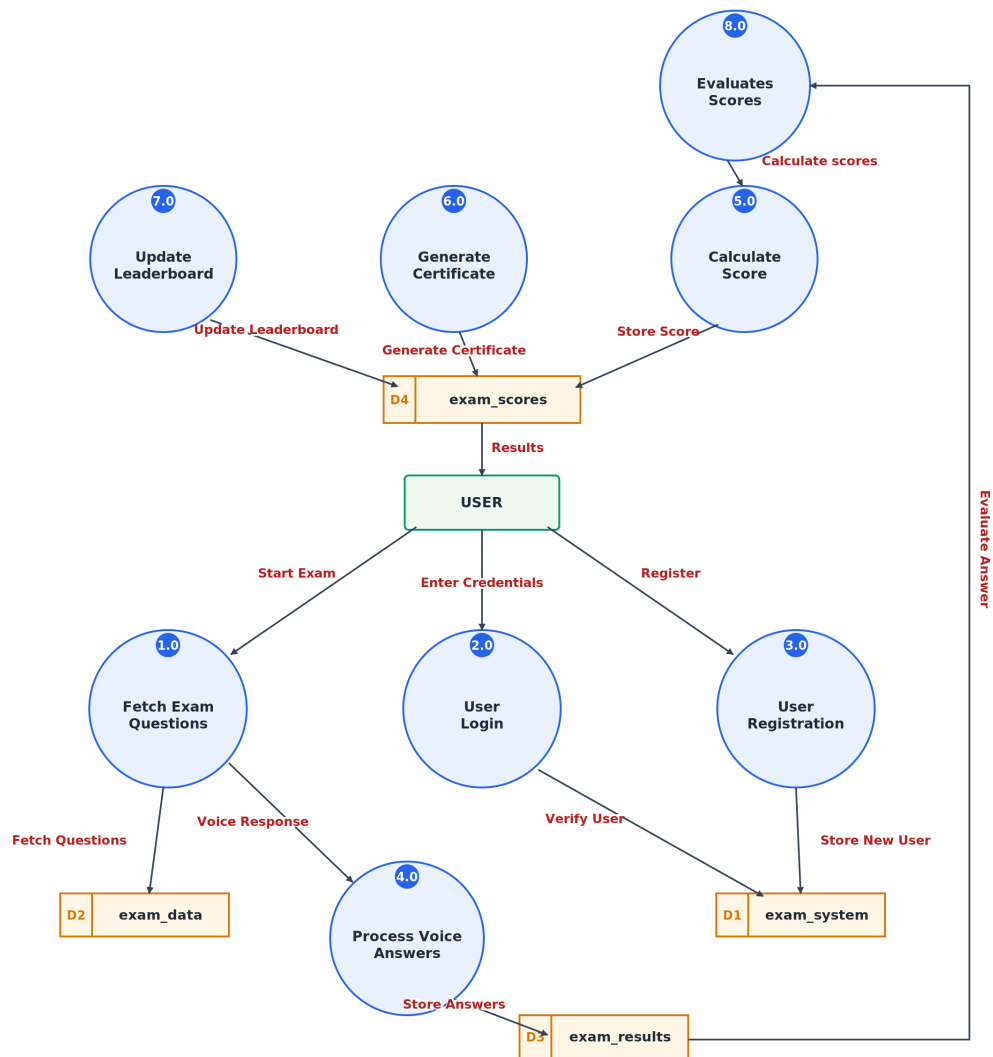


Figure 3: Level 2 Data Flow Diagram

10. Entity–Relationship Diagram

The ER model captures the data entities and the relationships between them. Four entities are maintained: **Users**, **Scores**, **Questions** and **exam_results**. A user *has* scores (one-to-many); scores relate to the questions the candidate *takes* (many-to-many); and questions are *evaluated* into exam_results (many-to-one).

Entity-Relationship Diagram

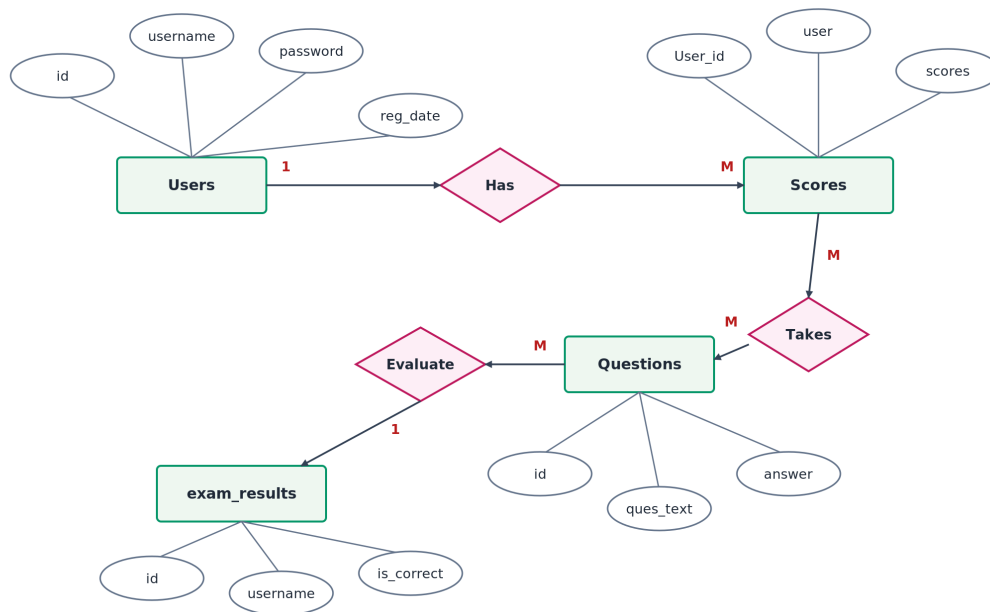


Figure 4: Entity-Relationship Diagram

11. Database Design (Data Dictionary)

The relational schema is derived directly from the ER model. The principal tables and their columns are documented below.

11.1 Table: users

Field	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique user identifier
username	VARCHAR(50)	NOT NULL, UNIQUE	Login name of the candidate
password	VARCHAR(255)	NOT NULL	Securely hashed password
reg_date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Date of registration

11.2 Table: questions

Field	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique question identifier
ques_text	TEXT	NOT NULL	Question text read aloud to the candidate
answer	VARCHAR(255)	NOT NULL	Expected correct answer

11.3 Table: scores

Field	Type	Constraints	Description
user_id	INT	FK → users.id	Reference to the candidate
user	VARCHAR(50)	NOT NULL	Candidate username
scores	INT	DEFAULT 0	Total score obtained by the candidate

11.4 Table: exam_results

Field	Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique result-row identifier
username	VARCHAR(50)	FK → users.username	Reference to the candidate
is_correct	TINYINT(1)	0 / 1	Whether the submitted answer was correct

11.5 Data Store Mapping

The data stores that appear in the DFDs map onto the physical tables as follows.

Data store (DFD)	Physical table	Contents
exam_system	users	Registered user credentials and profile
exam_data	questions	Question bank read during the exam
exam_results	exam_results	Per-question correctness log
exam_scores	scores	Aggregated candidate scores / leaderboard

12. Module Description

Module	Responsibility
Registration & Authentication	Handles new-user sign-up with duplicate-username checks and secure, hashed passwords, and authenticates returning users, establishing a role-aware session for candidate or admin.
Voice-Assisted Exam Engine	Fetches questions from the database, reads each one aloud using text-to-speech, and captures the candidate's spoken response through speech recognition, transcribing it to text for evaluation.
Question Management (Admin)	Lets the administrator create, update and delete questions in the exam_data store and organise the question bank used by the exam engine.
Answer Evaluation	Compares the transcribed answer with the stored answer using text-similarity / fuzzy matching and flags each response as correct or incorrect (is_correct).
Result Processing & Scoring	Aggregates per-question outcomes, computes the total score, and persists results to exam_results and scores for the candidate.
Certificate & Leaderboard	Generates a completion certificate for candidates who meet the pass threshold and updates the leaderboard so candidates can be ranked by score.
Admin Management	Provides administrators with oversight of user accounts and the ability to review stored results and question data.

13. Implementation Workflow

A typical end-to-end candidate session proceeds as follows:

- **Register / Login:** the candidate creates an account or signs in; credentials are verified against the users table.
- **Start Exam:** the exam engine fetches questions from exam_data and reads the first question aloud.
- **Answer by voice:** the candidate speaks the answer; the browser transcribes it and sends it to the server.
- **Evaluate:** the answer is compared with the stored answer and marked correct or incorrect.
- **Score & store:** the score is accumulated and per-question results are written to exam_results and scores.
- **Complete:** a certificate is generated, the leaderboard is updated, and the result is returned to the candidate.

14. Testing

The system was validated with functional test cases covering registration, authentication, the voice exam flow, evaluation, scoring and administration. Representative cases are listed below.

ID	Test Case	Expected Result	Status
TC01	Register new user with valid details	Account created; redirected to login	Pass
TC02	Register with an existing username	Rejected with 'username already exists'	Pass
TC03	Login with correct credentials	Access granted; dashboard shown	Pass
TC04	Login with wrong password	Access denied with an error message	Pass
TC05	Question read aloud on exam start	Question spoken via text-to-speech	Pass
TC06	Capture spoken answer	Answer transcribed and recorded	Pass
TC07	Evaluate a correct answer	is_correct = 1 and score increased	Pass
TC08	Evaluate a wrong answer	is_correct = 0; score unchanged	Pass
TC09	Compute final score	Total score calculated and stored	Pass
TC10	Generate certificate on pass	Certificate produced for the candidate	Pass
TC11	Update leaderboard after exam	Candidate rank reflected on leaderboard	Pass
TC12	Admin adds a new question	Question saved to the exam_data store	Pass

15. Results and Discussion

The implemented system successfully conducts an examination entirely through voice. Questions are delivered as speech, answers are captured and transcribed reliably in a quiet environment, and evaluation with text-similarity matching produces objective, repeatable scores. Candidates receive their results, certificate and leaderboard position immediately after finishing, eliminating manual grading effort. The role separation between candidate and administrator keeps the question bank and user data secure. In noisy conditions or with strong accents, transcription accuracy can drop; this is discussed as a limitation and a

target for future work.

16. Conclusion and Future Enhancements

The AI Voice-Assisted Exam System demonstrates how speech technology and lightweight natural-language evaluation can be combined with a standard web-and-database stack to deliver an accessible, automated examination platform. It meets its objectives of voice-based delivery, automatic answer evaluation, instant scoring, certification and leaderboard ranking, backed by a normalized relational database.

Future enhancements include:

- Support for descriptive answers using semantic (embedding-based) similarity rather than string matching.
- Multi-language voice support for questions and answers.
- AI-based proctoring (face and identity verification) to strengthen exam integrity.
- An analytics dashboard for administrators covering item difficulty and candidate performance.
- Adaptive testing that adjusts question difficulty to the candidate's responses.
- Noise-robust speech recognition to improve accuracy in real-world conditions.

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